CSA 0919 Programming in Java for Spring

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EASY LEVEL PROGRAM

1.

Programiz
Online Java Compiler

ReverseString.java

```java
import java.util.Scanner;

public class ReverseString {
    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);
        System.out.print("Enter String: ");
        String str = sc.nextLine();

        String rev = "";

        for (int i = str.length() - 1; i >= 0; i--) {
            rev = rev + str.charAt(i);
        }

        System.out.println("Reverse String: " + rev);
    }
}
```

Output

```
Enter String: ROSE
Reverse String: ESOR

=== Code Execution Successful ===
```

2

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StringToInteger.java

```java
        if(str.charAt(0) == '-') {
            sign = -1;
            i = 1;
        }

        for (; i < str.length(); i++) {
            char ch = str.charAt(i);

            if (ch >= '0' && ch <= '9') {
                num = num * 10 + (ch - '0');
            } else {
                valid = false;
                break;
            }
        }

        if (valid)
            System.out.println("Output Integer: " + (num * sign
                ));
        else
            System.out.println("Invalid Input");
    }
}
```

Output

```
Enter String: 1234
Output Integer: 1234

=== Code Execution Successful ===
```

3

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UsernameCheck.java

```java
import java.util.Scanner;

public class UsernameCheck {
    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);

        System.out.print("Enter the user name: ");
        String name1 = sc.nextLine();

        System.out.print("Reenter the user name: ");
        String name2 = sc.nextLine();

        if(name1.equals(name2))
            System.out.println("User name is Valid");
        else
            System.out.println("User name is Invalid");
    }
}
```

Output

```
Enter the user name: ROSE
Reenter the user name: ROSE
User name is Valid

=== Code Execution Successful ===
```

4

Programiz
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SortNames.java

```java
        for(int i=0;i<names.length;i++){
            for(int j=i+1;j<names.length;j++){

                if(order=='A'){
                    if(names[i].compareTo(names[j])>0){
                        String temp=names[i];
                        names[i]=names[j];
                        names[j]=temp;
                    }
                }
                else{
                    if(names[i].compareTo(names[j])<0){
                        String temp=names[i];
                        names[i]=names[j];
                        names[j]=temp;
                    }
                }
            }
        }
        for(String n:names)
            System.out.println(n);
    }
```

Output

```
Order (A/D): A
Apple
Banana
Carrot
Jack
Radish

=== Code Execution Successful ===
```

5

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SpecialCharCount.java

```java
    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);
        System.out.print("Enter line: ");
        String str = sc.nextLine();

        int count = 0;

        System.out.println("Special Characters:");

        for(int i=0;i<str.length();i++){
            char ch = str.charAt(i);

            if(!Character.isLetterOrDigit(ch) && !Character
                .isSpaceChar(ch)){
                System.out.print(ch + " ");
                count++;
            }
        }

        System.out.println("\nNumber of Special Characters: " +
            count);
    }
```

Output

```
Enter line: i am rose
Special Characters:

Number of Special Characters: 0

=== Code Execution Successful ===
```

6

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CountVowels.java

```java
import java.util.Scanner;

public class CountVowels {
    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);
        System.out.print("Enter Statement: ");
        String str = sc.nextLine();

        int count = 0;

        for(int i = 0; i < str.length(); i++) {
            char ch = Character.toLowerCase(str.charAt(i));

            if(ch=='a' || ch=='e' || ch=='i' || ch=='o' || ch
                =='u') {
                count++;
            }
        }

        System.out.println("Number of vowels = " + count);
    }
}
```

Output

```
Enter Statement: India is my country
Number of vowels = 6

=== Code Execution Successful ===
```

7

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VowelConsonantSeparate.java

```java

public class VowelConsonantSeparate {
    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);
        System.out.print("Enter Word: ");
        String str = sc.nextLine().toLowerCase();

        System.out.print("Vowels: ");
        for(int i=0;i<str.length();i++){
            char ch=str.charAt(i);
            if(ch=='a'||ch=='e'||ch=='i'||ch=='o'||ch=='u')
                System.out.print(ch+" ");
        }

        System.out.print("\nConsonants: ");
        for(int i=0;i<str.length();i++){
            char ch=str.charAt(i);
            if(ch>='a' && ch<='z' &&
               !(ch=='a'||ch=='e'||ch=='i'||ch=='o'||ch=='u'))
                System.out.print(ch+" ");
        }
    }
}
```

Output

```
Enter Word: MEDIAN
Vowels: e i a
Consonants: m d n
=== Code Execution Successful ===
```

8

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SearchCharacter.java

```java
        Scanner sc = new Scanner(System.in);

        System.out.print("Enter String: ");
        String str = sc.nextLine();

        System.out.print("Enter character to search: ");
        char ch = sc.next().charAt(0);

        boolean found = false;

        for(int i=0;i<str.length();i++){
            if(str.charAt(i)==ch){
                System.out.println(ch + " found at index: " + i
                    );
                found = true;
                break;
            }
        }

        if(!found)
            System.out.println("Character not found");
    }
}
```

Output

```
Enter String: Hello World

z
Enter character to search: Character not found

=== Code Execution Successful ===
```

9

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ReverseAlphabetOrder.java

```java
public class ReverseAlphabetOrder {
    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);
        System.out.print("Enter Word: ");
        String str = sc.nextLine().toUpperCase();

        char arr[] = str.toCharArray();

        for(int i=0;i<arr.length;i++){
            for(int j=i+1;j<arr.length;j++){
                if(arr[i] < arr[j]){
                    char temp = arr[i];
                    arr[i] = arr[j];
                    arr[j] = temp;
                }
            }
        }

        System.out.print("Alphabetical Order: ");
        for(char c:arr)
            System.out.print(c + " ");
    }
}
```

Output

```
Enter Word: MANIPULATION
Alphabetical Order: U T P O N N M L I I A A
=== Code Execution Successful ===
```

10

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RemoveVowels.java

Share Run

```java
import java.util.Scanner;

public class RemoveVowels {
    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);
        System.out.print("Enter String: ");
        String str = sc.nextLine();

        String result = "";

        for(int i=0;i<str.length();i++){
            char ch = Character.toLowerCase(str.charAt(i));

            if(!(ch=='a'||ch=='e'||ch=='i'||ch=='o'||ch=='u')){
                result = result + str.charAt(i);
            }
        }

        System.out.println("String without vowels: " + result);
    }
}
```

Output

Clear

```
Enter String: EDUCATION
String without vowels: DCTN

=== Code Execution Successful ===
```

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MatrixMultiplication.java

Share Run

```java

        int a[][] = {{1,2},{5,3}};
        int b[][] = {{2,3},{4,1}};
        int c[][] = new int[2][2];

        for(int i=0;i<2;i++){
            for(int j=0;j<2;j++){
                c[i][j]=0;
                for(int k=0;k<2;k++){
                    c[i][j] += a[i][k] * b[k][j];
                }
            }
        }

        System.out.println("Matrix Multiplication Result:");
        for(int i=0;i<2;i++){
            for(int j=0;j<2;j++){
                System.out.print(c[i][j]+" ");
            }
            System.out.println();
        }
    }
```

Output

Clear

```
Matrix Multiplication Result:
10 5
22 18

=== Code Execution Successful ===
```

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MatrixAddition.java

Share Run

```java
public class MatrixAddition {
    public static void main(String[] args) {

        int a[][] = {{1,2},{5,3}};
        int b[][] = {{2,3},{4,1}};
        int c[][] = new int[2][2];

        for(int i=0;i<2;i++){
            for(int j=0;j<2;j++){
                c[i][j] = a[i][j] + b[i][j];
            }
        }

        System.out.println("Matrix Addition Result:");
        for(int i=0;i<2;i++){
            for(int j=0;j<2;j++){
                System.out.print(c[i][j]+" ");
            }
            System.out.println();
        }
    }
}
```

Output

Clear

```
Matrix Addition Result:
3 5
9 4

=== Code Execution Successful ===
```

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MergeArrays.java

Share Run

```java
        int arr1[] = {1,3,4,5};
        int arr2[] = {2,4,6,8};

        ArrayList<Integer> list = new ArrayList<>();

        int i=0,j=0;

        while(i<arr1.length && j<arr2.length){
            if(arr1[i] < arr2[j])
                list.add(arr1[i++]);
            else
                list.add(arr2[j++]);
        }

        while(i<arr1.length)
            list.add(arr1[i++]);

        while(j<arr2.length)
            list.add(arr2[j++]);

        System.out.println("Merged Array: "+list);
    }
}
```

Output Clear

```
Merged Array: [1, 2, 3, 4, 4, 5, 6, 8]

=== Code Execution Successful ===
```

14

MeanMedianMode.java

Share Run

```java
        else
            median = arr[n/2];

        // Mode
        int mode = arr[0], maxCount=0;

        for(int i=0;i<n;i++){
            int count=0;
            for(int j=0;j<n;j++){
                if(arr[i]==arr[j])
                    count++;
            }
            if(count>maxCount){
                maxCount=count;
                mode=arr[i];
            }
        }

        System.out.println("Mean = "+mean);
        System.out.println("Median = "+median);
        System.out.println("Mode = "+mode);
    }
}
```

Output Clear

```
Mean = 20.0
Median = 19.0
Mode = 16

=== Code Execution Successful ===
```

15

CompositeCount.java

Share Run

```java
        int arr[] = {16,18,27,16,23,21,19};
        int count = 0;

        for(int i=0;i<arr.length;i++){
            int num = arr[i];

            if(num>1){
                int factors=0;

                for(int j=1;j<=num;j++){
                    if(num%j==0)
                        factors++;
                }

                if(factors>2)
                    count++;
            }
        }

        System.out.println("Number of Composite Numbers = "
            +count);
    }
```

Output Clear

```
Number of Composite Numbers = 5

=== Code Execution Successful ===
```

16

RightTriangle.java

```java
import java.util.Scanner;

public class RightTriangle {
    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);
        System.out.print("Enter n: ");
        int n = sc.nextInt();

        for(int i=1;i<=n;i++){
            for(int j=1;j<=n-i;j++)
                System.out.print("  ");

            for(int k=1;k<=i;k++)
                System.out.print("* ");

            System.out.println();
        }
    }
}
```

Output

```
Enter n: 4
      *
    * *
  * * *
* * * *

=== Code Execution Successful ===
```

17

PascalPattern.java

```java
import java.util.Scanner;

public class PascalPattern {
    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);
        System.out.print("Enter rows: ");
        int n = sc.nextInt();

        for(int i=0;i<n;i++){
            int num=1;

            for(int space=0;space<n-i;space++)
                System.out.print(" ");

            for(int j=0;j<=i;j++){
                System.out.print(num+" ");
                num = num*(i-j)/(j+1);
            }
            System.out.println();
        }
    }
}
```

Output

```
Enter rows: 5
     1
    1 1
   1 2 1
  1 3 3 1
 1 4 6 4 1

=== Code Execution Successful ===
```

18

RectanglePattern.java

```java
import java.util.Scanner;

public class RectanglePattern {
    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);

        System.out.print("Enter symbol: ");
        char ch = sc.next().charAt(0);

        System.out.print("Enter rows: ");
        int rows = sc.nextInt();

        System.out.print("Enter columns: ");
        int cols = sc.nextInt();

        for(int i=1;i<=rows;i++){
            for(int j=1;j<=cols;j++){
                System.out.print(ch+" ");
            }
            System.out.println();
        }
    }
}
```

Output

```
Enter symbol: #
Enter rows: 3
Enter columns: 5
# # # # #
# # # # #
# # # # #

=== Code Execution Successful ===
```

19

NumberDiamond.java

```java
public class NumberDiamond {
    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);

        System.out.print("Enter number: ");
        int num = sc.nextInt();

        System.out.print("Max times: ");
        int n = sc.nextInt();

        for(int i=1;i<=n;i++){
            for(int j=1;j<=i;j++)
                System.out.print(num);
            System.out.println();
        }

        for(int i=n-1;i>=1;i--){
            for(int j=1;j<=i;j++)
                System.out.print(num);
            System.out.println();
        }
    }
}
```

Output

```
Enter number: 1
Max times: 3
1
11
111
11
1

=== Code Execution Successful ===
```

20

InvertedPyramid.java

```java
import java.util.Scanner;

public class InvertedPyramid {
    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);
        System.out.print("Enter rows: ");
        int n = sc.nextInt();

        for(int i=n;i>=1;i--){
            for(int space=1;space<=n-i;space++)
                System.out.print(" ");

            for(int j=1;j<=2*i-1;j++)
                System.out.print("*");

            System.out.println();
        }
    }
}
```

Output

```
Enter rows: 3
*****
 ***
  *

=== Code Execution Successful ===
```

21

CharacterTriangle.java

```java
import java.util.Scanner;

public class CharacterTriangle {
    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);

        System.out.print("Enter character: ");
        char ch = sc.next().charAt(0);

        System.out.print("Max times: ");
        int n = sc.nextInt();

        for(int i=1;i<=n;i++){
            for(int j=1;j<=i;j++){
                System.out.print(ch + " ");
            }
            System.out.println();
        }
    }
}
```

Output

```
Enter character: %
Max times: 3
%
% %
% % %

=== Code Execution Successful ===
```

22

HollowSquare.java

```java
import java.util.Scanner;

public class HollowSquare {
    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);

        System.out.print("Enter symbol: ");
        char ch = sc.next().charAt(0);

        System.out.print("Enter size: ");
        int n = sc.nextInt();

        for(int i=1;i<=n;i++){
            for(int j=1;j<=n;j++){
                if(i==1 || i==n || j==1 || j==n)
                    System.out.print(ch + " ");
                else
                    System.out.print("  ");
            }
            System.out.println();
        }
```

Output

```
Enter symbol: $
Enter size: 4
$ $ $ $
$     $
$     $
$ $ $ $

=== Code Execution Successful ===
```

23

RepeatedNumber.java

```java
import java.util.Scanner;

public class RepeatedNumber {
    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);

        System.out.print("Enter rows: ");
        int n = sc.nextInt();

        for(int i=1;i<=n;i++){
            for(int j=1;j<=i;j++){
                System.out.print(i + " ");
            }
            System.out.println();
        }
    }
}
```

Output

```
Enter rows: 4
1
2 2
3 3 3
4 4 4 4

=== Code Execution Successful ===
```

24

SquarePattern.java

```java
import java.util.Scanner;

public class SquarePattern {
    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);

        System.out.print("Enter rows: ");
        int n = sc.nextInt();

        int num = 1;

        for(int i=1;i<=n;i++){
            for(int j=1;j<=i;j++){
                System.out.print((num*num) + " ");
                num++;
            }
            System.out.println();
        }
    }
}
```

Output

```
Enter rows: 4
1
4 9
16 25 36
49 64 81 100

=== Code Execution Successful ===
```

25

NumberDiamond2.java

```java

public class NumberDiamond2 {
    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);

        System.out.print("Enter rows: ");
        int n = sc.nextInt();

        for(int i=1;i<=n;i++){
            for(int j=1;j<=i;j++){
                System.out.print(i + " ");
            }
            System.out.println();
        }

        for(int i=n-1;i>=1;i--){
            for(int j=1;j<=i;j++){
                System.out.print(i + " ");
            }
            System.out.println();
        }
    }
}
```

Output

```
Enter rows: 4
1
2 2
3 3 3
4 4 4 4
3 3 3
2 2
1

=== Code Execution Successful ===
```

26

HollowDollar.java

```java
import java.util.Scanner;

public class HollowDollar {
    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);

        System.out.print("Enter size: ");
        int n = sc.nextInt();

        for(int i=1;i<=n;i++){
            for(int j=1;j<=n;j++){
                if(i==1 || i==n || j==1 || j==n)
                    System.out.print("$ ");
                else
                    System.out.print("  ");
            }
            System.out.println();
        }
    }
}
```

Output

```
Enter size: 3
$ $ $
$   $
$ $ $

=== Code Execution Successful ===
```

27

ReverseNumber.java

```java
    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);

        System.out.print("Enter number: ");
        String input = sc.nextLine();

        try {
            int num = Integer.parseInt(input);
            int rev = 0;

            while(num != 0){
                int digit = num % 10;
                rev = rev * 10 + digit;
                num = num / 10;
            }

            System.out.println("Reverse Number: " + rev);
        }
        catch(Exception e){
            System.out.println("Invalid Input");
        }
```

Output

```
Enter number: 14567
Reverse Number: 76541

=== Code Execution Successful ===
```

28

InvertedPyramid2.java

```java
import java.util.Scanner;

public class InvertedPyramid2 {
    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);

        System.out.print("Enter rows: ");
        int n = sc.nextInt();

        for(int i=n;i>=1;i--){
            for(int space=1;space<=n-i;space++)
                System.out.print(" ");

            for(int j=1;j<=2*i-1;j++)
                System.out.print("*");

            System.out.println();
        }
    }
}
```

Output

```
Enter rows: 3
*****
 ***
  *

=== Code Execution Successful ===
```

29

BinaryReverse.java

```java
            String binary = "";
            while(num > 0){
                binary = (num % 2) + binary;
                num = num / 2;
            }

            String rev = "";
            for(int i=binary.length()-1;i>=0;i--)
                rev += binary.charAt(i);

            int result = 0;
            for(int i=0;i<rev.length();i++){
                result = result*2 + (rev.charAt(i)-'0');
            }

            System.out.println("Output: " + result);
        }
        catch(Exception e){
            System.out.println("Invalid Input");
        }
    }
```

Output

```
Enter number: 11
Output: 13

=== Code Execution Successful ===
```

30

VotingEligibility.java

```java
public class VotingEligibility {
    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);

        System.out.print("Enter your age: ");
        String input = sc.nextLine();

        try {
            int age = Integer.parseInt(input);

            if(age >= 18)
                System.out.println("You are eligible to vote");
            else if(age >= 0)
                System.out.println("You are allowed to vote
                    after " + (18-age) + " years");
            else
                System.out.println("Invalid Age");
        }
        catch(Exception e){
            System.out.println("Invalid Input");
        }
    }
```

Output

```
Enter your age: 7
You are allowed to vote after 11 years

=== Code Execution Successful ===
```